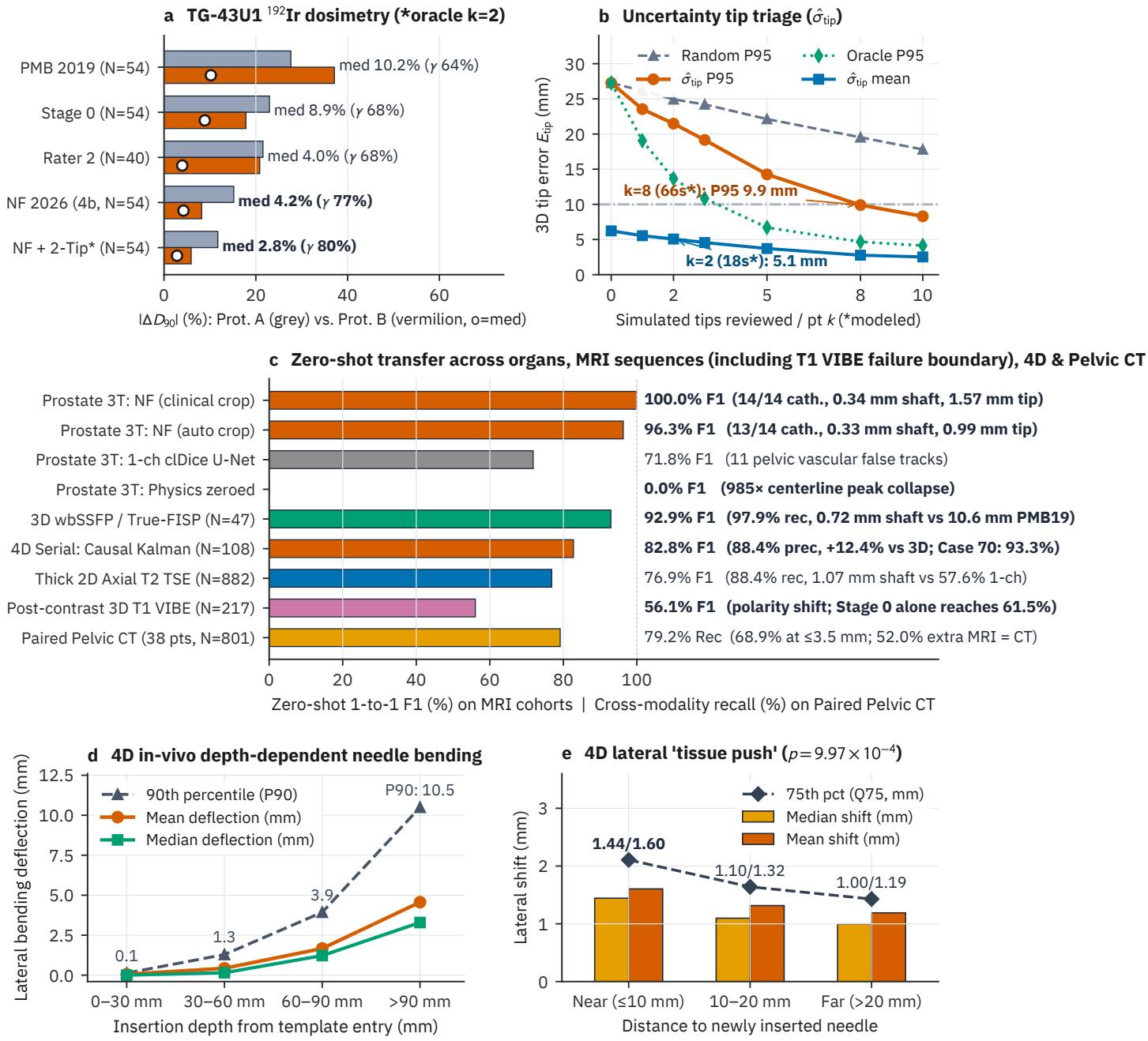
a TG-43U1 ^{192}Ir dosimetry (*oracle k=2)
PMB 2019 (N=54)
Stage 0 (N=54)
Rater 2 (N=40)
NF 2026 (4b, N=54)
NF + 2-Tip* (N=54)
med 10.2% (γ 64%)
med 8.9% (γ 68%)
med 4.0% (γ 68%)
med 4.2% (γ 77%)
med 2.8% (γ 80%)
0
20
40
60
|ΔD90| (%): Prot. A (grey) vs. Prot. B (vermilion, o=med)
b Uncertainty tip triage ($\hat{\sigma}_{tip}$)
Random P95
Oracle P95
$\hat{\sigma}_{tip}$ P95
$\hat{\sigma}_{tip}$ mean
3D tip error E_{tip} (mm)
k=8 (66s*): P95 9.9 mm
k=2 (18s*): 5.1 mm
0
2
5
8
10
Simulated tips reviewed / pt k (*modeled)
c Zero-shot transfer across organs, MRI sequences (including T1 VIBE failure boundary), 4D & Pelvic CT
Prostate 3T: NF (clinical crop)
Prostate 3T: NF (auto crop)
Prostate 3T: 1-ch clDice U-Net
Prostate 3T: Physics zeroed
3D wbSSFP / True-FISP (N=47)
4D Serial: Causal Kalman (N=108)
Thick 2D Axial T2 TSE (N=882)
Post-contrast 3D T1 VIBE (N=217)
Paired Pelvic CT (38 pts, N=801)
100.0% F1 (14/14 cath., 0.34 mm shaft, 1.57 mm tip)
96.3% F1 (13/14 cath., 0.33 mm shaft, 0.99 mm tip)
71.8% F1 (11 pelvic vascular false tracks)
0.0% F1 (985× centerline peak collapse)
92.9% F1 (97.9% rec, 0.72 mm shaft vs 10.6 mm PMB19)
82.8% F1 (88.4% prec, +12.4% vs 3D; Case 70: 93.3%)
76.9% F1 (88.4% rec, 1.07 mm shaft vs 57.6% 1-ch)
56.1% F1 (polarity shift; Stage 0 alone reaches 61.5%)
79.2% Rec (68.9% at ≤3.5 mm; 52.0% extra MRI = CT)
0
20
40
60
80
100
Zero-shot 1-to-1 F1 (%) on MRI cohorts | Cross-modality recall (%) on Paired Pelvic CT
d 4D in-vivo depth-dependent needle bending
90th percentile (P90)
Mean deflection (mm)
Median deflection (mm)
P90: 10.5
3.9
1.3
0.1
Lateral bending deflection (mm)
0–30 mm
30–60 mm
60–90 mm
>90 mm
Insertion depth from template entry (mm)
e 4D lateral 'tissue push' ($p = 9.97 \times 10^{-4}$)
75th pct (Q75, mm)
Median shift (mm)
Mean shift (mm)
1.44/1.60
1.10/1.32
1.00/1.19
Lateral shift (mm)
Near (≤10 mm)
10–20 mm
Far (>20 mm)
Distance to newly inserted needle